



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS-343: Web Engineering

Class: BSAI-01

Lab 08: Media queries and Responsive web design

Date: March 27, 2026

Muhammad Sikandar Hussain

502808



Lab 08: Media queries and Responsive web design

Introduction:

Students have learned advanced concepts of CSS3, media queries, and responsive web design. In this lab, student will practically design a responsive web page using the concepts learned in the lectures.

Lab Objectives:

The objective of this lab is to help students in designing a responsive web page using divisions, CSS3, and media queries. Students will apply different CSS styles and responsive web designing concepts they have learned in the lectures to design the layout of a web page.

Tools:

Dreamweaver, notepad, browser.

Helping Material:

Lecture slides.

W3Schools: https://www.w3schools.com/html/html_responsive.asp

https://www.w3schools.com/Css/css_rwd_intro.asp

Plenty of DIV examples:

http://www.mobilefish.com/tutorials/css/css_quickguide_div_examples_containers.html

Lab Task

Task 1

In the previous labs, you have designed a website which is similar to BBC's website using tables and divisions. Designing the layout of a website using tables and divisions alone is non-trivial due to limited flexibility, layout distortions, and an inability to response against the change in screen size (e.g. browser's window resized). Professional websites like BBC are designed using HTML5 and CSS3. The responsiveness of the BBC website can be observed by resizing browser's window. When carefully observing this layout, one can easily see that there are three different layouts which are adapted whenever the screen size is changed. The three layouts are: a) smart phone layout, b) tablet layout, and c) desktop layout.



Lab Task:

Design a responsive web page which has a layout similar to the BBC's website. Your layout must be based on DIVs and CSS3. There must be three different layouts for the smart phone, tablet, and desktop. The layouts must be adapted automatically upon a relevant change in the screen size. An RWD example is already uploaded on LMS. You should use it as an example and extend it to design a responsive web page which is similar to BBC's web site.

Hints

1. Feel free to browse BBC's website and look into its page source to see how the layout is designed.
2. You can change the color scheme as per your choice.
3. BBC's website has a menu (More) and a search bar. You can skip both in your layout.
4. You are encouraged to use most of the HTML tags and CSS styles you have learned so far or even new tags, which you can learn from W3Schools.
5. Make good use of HTML colors, fonts, font-family, font-sizes and other styles for the **look and feel** of the web page.
6. Make sure to test your website with different resolutions by changing the resolution of your screen.



NEWS

[Sign In](#)[Home](#) [News](#) [Sport](#) [Business](#) [Culture](#) [Travel](#)

Welcome to today's updates

Friday, 27 March 2026

TOP STORY

Urban Development Initiative Aims for Sustainable Growth

A comprehensive plan for city expansion includes green spaces, efficient public transport, and eco-friendly housing to accommodate growing population needs.

[Read full story](#)

ECONOMY

Stock Markets Show Positive Trends Amid Global Recovery

Economic indicators point to improved investor confidence this quarter.

INNOVATION

Researchers Develop Affordable Smart Agriculture Tools

The new system enables farmers to optimize crop yields with minimal resources.

GLOBAL

International Health Campaign Reaches Remote Communities

Medical teams provide essential services to underserved areas worldwide.

ENVIRONMENT

Youth-Led Conservation Project Gains Momentum

Students organize reforestation efforts across educational institutions.

News

Modernized Community Center Unveils Advanced Learning Facilities

The renovated space features interactive classrooms, digital resources, and collaborative areas designed for contemporary education needs.

Entrepreneurs Showcase Cutting-Edge Solutions at Tech Showcase

Attendees explored innovative projects spanning healthcare, transportation, education, and entrepreneurial development.

Weather Advisory Issued for Approaching Storm System

Residents are encouraged to prepare for potential heavy rainfall and take precautions during the forecasted weather event.

Sport

Team Captain Credits Collective Performance for Victory

The leader emphasized strategic play, resilience, and precise execution as key factors in securing the championship title.

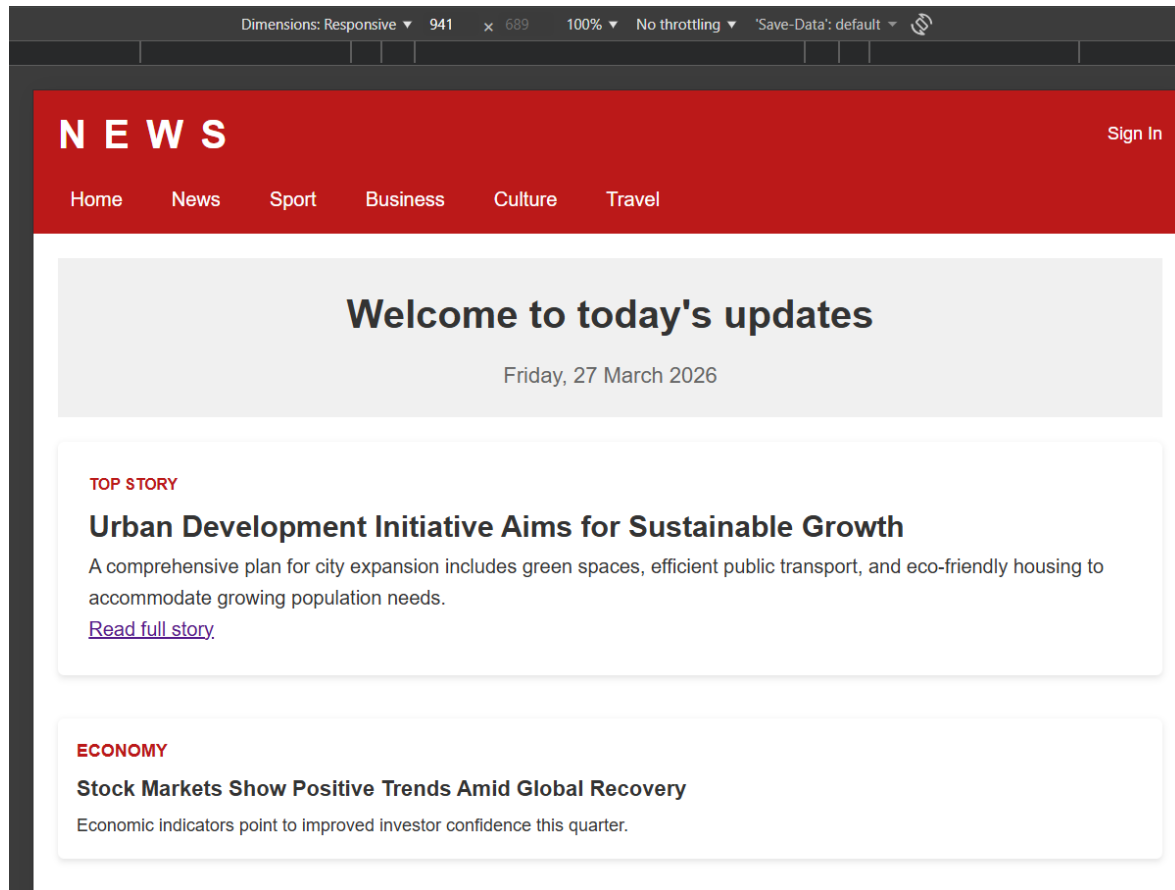
Inter-School Tournament Attracts Enthusiastic Spectators

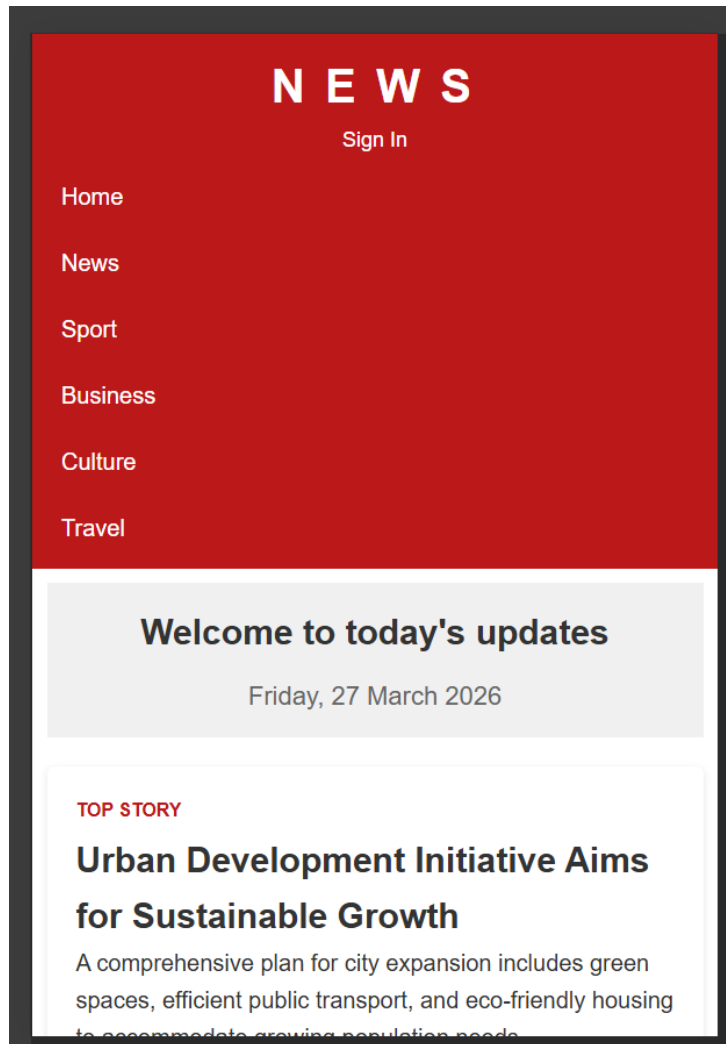
Supporters, educators, and participants created an electric atmosphere for the highly anticipated final confrontation.

Campus Wellness Initiative Promotes Active Lifestyles

Coordinators report that regular physical activity enhances student vitality and academic engagement.

Responsive news page for Web Engineering Lab 08





Deliverables

- ☐ Each submission is individual with the following composition:
 - Description document
 - Source Code (Web Pages)
- ☐ Convert your submission files into a zip folder and name it as given below, finally upload the zip folder to LMS.
 - Name – Registration No. – Section



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science